

Sangryul Kim

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Summary

I am a master’s student at KAIST AI Graduate School in the xfact lab under Professor James Thorne, researching NLP, text retrieval, and deep learning. I enjoy creating tech products, with a focus on robust idea and developing MVPs. My work involves using large language models (LLMs) for search, retrieval, and SQL generation, and I am exploring efficient learning and data compression techniques. I have experience in full fine-tuning large models, building RAG tools with Korean models, and deploying models with Hugging Face, as well as skills in deep learning, data engineering, and web development.

Education

MS	KAIST AI - GPA: 3.94/4.3 - Advisor: James Thorne / Explainable Factual Reasoning Lab(xfact) 🔗	Mar. 2023 to Feb. 2025(planned)
BS	Chung-Ang University , Major : Computer Science and Engineering / Minor : Applied Statistics GPA: 3.98/4.5	Mar. 2016 to Feb. 2023

Publications

Context Filtering with Reward Modeling in Question Answering In Proceedings of the 31st International Conference on Computational Linguistics (COLING 2025) <u>Sangryul Kim</u> , James Thorne	Jan. 2025 (TBA)
ProbGate at EHRSQL 2024: Enhancing SQL Query Generation Accuracy through Probabilistic Threshold Filtering and Error Handling The 6th Clinical Natural Language Processing Workshop at NAACL 2024 <u>Sangryul Kim</u> , Donghee Han, Sehyun Kim	May 2024
Re3val: Reinforced and Reranked Generative Retrieval Findings of the Association for Computational Linguistics: EACL 2024 Euiyul Song, <u>Sangryul Kim</u> , Haeju Lee, Joonkee Kim, James Thorne	Mar. 2024
Implementing chatbot from K-drama:learning how to speak from the main characters Korea Computer Congress 2022 (KCC 2022, Domestic Conference) <u>Sangryul Kim</u> , Seunguk Yu, Joonghoon Kim, Yoon-Sik Cho	Jun. 2022

Experience

NAVER WEBTOON , Machine Learning Engineer, Intern - Developed an end-to-end program to block inappropriate ads by analyzing and visualizing data from ad images. - Work Experience Field : Data Preprocessing, Statistical Machine Learning, Deploy and Manage the data and model	Seongnam-si, Republic of Korea Jul. 2022 to Oct. 2022
Cyber Operations Command of Republic of Korea , Discharged to the rank of sergeant Developed and operated military training program for cyber attack simulation with C++ and Windows API.	Seoul, Republic of Korea Feb. 2019 to Sep. 2020

- Developed and operated malware and C2 server with C++, Python and Mimikatz.
- Developed a system for mobile security diagnosis using Vue.js, Node.js, ADB and MetaDefender.

ONYCOM Inc, Software Engineer

Seoul, Republic of Korea
Jan. 2018 to Jan. 2019

- Developed the pre-release version of [IMQA](#) MPM and Crash service as a Web programmer.

Projects

Gompada / [github](#)

2022

- Developed "Gompada," an AI grading tool for descriptive answers using SBert and FastText, focusing on keyword matching and contextual similarity.
- Designed and developed front-end (Vue.js) and back-end (FastAPI) services, optimizing ML model loading for faster response times.

Help Diana / [github](#) / [demo](#)

2021

- Developed a web service to bridge the information gap between doctors and patients using AI, Vue.js PWA, Spring, MySQL, AWS, and Docker.
- Led project planning, design, front-end development, AWS setup, data crawling, and API integration, including OCR and medical word tokenization.

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2021

- Developed a stock trading prediction project using FastAI, TensorFlow, PyTorch, FastAPI, and Docker.
- Led planning, model design, server development, data collection, model experimentation, and deploying models for stock prediction using CNN and LSTM, optimizing performance for real-time investment strategies.

Additional Experience

CIT Gangnam Apgujeong Coding Academy Instructor(Sep. 2021 to Mar. 2022): Worked as a coding academy instructor, prepares for overseas universities, and conducts Python, Data Science, Machine Learning classes and projects for adults

Certificated Boostcamp AI Tech 3th by Naver Connect Cooperation(Jan. 2022 to June 2022): Participated in NLP track, Got 1st place in the first team image classification competition on stages.ai, Experienced Text Retrieval, NER Downstream Task of NLP

Certificated Google Machine Learning Bootcamp by Google Developers(Aug. 2021 to Nov. 2021): Certificated Deep Learning Specialization(Coursera) by Andrew Ng, Participated in Kaggle Tabular Competition

Certificated SW Maestro 8th by Ministry of Science and ICT(Aug. 2017 to Dec. 2017): Developed "Mononity", Perform mobile performance data extraction and analysis, Do extraction and diagnosis of Android smartphone performance information using ADB Shell

Awards

2nd Place: 2023 AI Graduate School Challenge hosted by Korea Ministry of Science and ICT / [github](#)

- Present "Explainable and Customizable Multihop Question Answering", Developed a RAG system with various custom features and integrated Korean language models.

Excellence Thesis Award: 2017 FALL KSII Conference

- "Analysis of Emotion Prediction Model by Genre Using Musical Features" / [Sangryul Kim](#), Hyunki Lim. Dae-Won Kim